

LABORATORY OBSERVATION / RECORD

Full Stack Web Development | Course Code: 23CM4121 | Observation Task No. 1

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Design and develop a structured, multi-section web page — **Bookify Library Portal** — using the fundamental building blocks of HTML and HTML5.

To understand and implement the fundamental building blocks of HTML and HTML5 by constructing a functional library-membership web page that demonstrates semantic layout, rich text formatting, navigation, tabular data, form controls, and multimedia embedding.

- To study and apply HTML5 semantic structure elements — `<header>`, `<nav>`, `<article>`, `<section>`, `<footer>` — to build a page skeleton.
- To format textual content using `<sup>` and `<sub>` tags along with descriptive paragraphs.
- To build two-tier navigation using both `<ol>` and `<ul>` list structures with anchor-based bookmarks and external links.
- To organise a book-catalogue dataset using `<table>`, `<thead>`, and `<tbody>`.
- To design a comprehensive membership-registration form using HTML5 input types such as `color`, `range`, `number`, and `date`, including client-side required validation.
- To embed native `<audio>` and `<video>` elements with playback controls.

HTML (HyperText Markup Language) is the foundational markup language of the web, describing the structure of a document through a system of nested elements. HTML5 extended this vocabulary with semantic elements that convey meaning about the role of a block of content, rather than relying purely on generic `<div>` containers. This improves accessibility, search-engine indexing, and long-term maintainability of markup.

4.1 Semantic Layout Elements

Elements such as `<header>`, `<nav>`, `<article>`, `<section>`, and `<footer>` divide a page into meaningful regions: introductory content, navigation links, self-contained editorial content, grouped thematic content, and closing information, respectively.

4.2 Text-Level Semantics

The `<sup>` and `<sub>` elements render inline text above or below the baseline, commonly used for chemical formulae, footnote markers, or ordinal indicators.

4.3 Tabular Data

The `<table>` element, combined with `<thead>` and `<tbody>`, separates column headings from row data, allowing browsers and assistive technology to interpret rows and columns correctly.

4.4 HTML5 Form Controls

HTML5 introduced specialised input types — `color`, `range`, `number`, and `date` — that provide native UI widgets (colour pickers, sliders, steppers, and date pickers) along with built-in constraint validation attributes such as `required`, `min`, `max`, and `minlength`.

4.5 Native Multimedia

The `<audio>` and `<video>` elements allow browsers to play media files natively using the `controls` attribute, removing the need for third-party plugins such as Flash.

1. Create the project directory structure: `Observation/ObTask1/html` and `Observation/ObTask1/media`.
2. Create `library.html` and lay out the page skeleton using semantic containers.
3. Build the page header with the site title, a superscript/subscript styled tagline, and an introductory paragraph describing the portal.
4. Construct a two-column navigation bar containing an ordered list of dashboard links and an unordered list of external/portal links, with one internal anchor bookmark.
5. Place the library logo beside a descriptive article about the organisation's history.

- Build a book-catalogue table listing Book ID, Title, and Status inside thead/tbody.
- Implement a membership-registration form inside a fieldset, combining text, email, password, radio, checkbox, select, color, range, number, and date controls, with required validation on key fields.
- Embed native <audio> and <video> elements with the controls attribute enabled.
- Render the page in Google Chrome (via a Playwright-driven headless Chromium instance) and verify every element and test case listed in Section 8.

File: library.html

```
<section>
  <div id="catalogue">
    <table>
      <thead>
        <tr><th>Book ID</th><th>Title</th><th>Status</th></tr>
      </thead>
      <tbody>
        <tr><td>BK-1042</td><td>The Silent Orchard</td><td>Available</td></tr>
        <tr><td>BK-2087</td><td>Vectors of Tomorrow</td><td>On Loan</td></tr>
        <tr><td>BK-3391</td><td>Algorithms in the Attic</td><td>Reserved</td></tr>
      </tbody>
    </table>
  </div>
</section>

<div class="formbox" id="registerForm">
  <fieldset>
    <legend>New Member Registration</legend>
    <form action="#" method="post">
      <label for="uname">Full Name</label>
      <input type="text" id="uname" name="uname" placeholder="Enter full name" required><br>

      <label for="uemail">Email Address</label>
      <input type="email" id="uemail" name="uemail" placeholder="you@example.com" required><br>

      <label for="upass">Create Password</label>
      <input type="password" id="upass" name="upass" minlength="6"><br>

      <label>Membership Type</label>
      <input type="radio" id="mtype1" name="mtype" checked><label for="mtype1" style="width:auto;">Standard</label>
      <input type="radio" id="mtype2" name="mtype"><label for="mtype2" style="width:auto;">Premium</label><br>

      <label>Reading Interests</label>
      <input type="checkbox" id="int1" name="interest"><label for="int1" style="width:auto;">Fiction</label>
      <input type="checkbox" id="int2" name="interest"><label for="int2" style="width:auto;">Science</label>
      <input type="checkbox" id="int3" name="interest"><label for="int3" style="width:auto;">History</label>
      <input type="checkbox" id="int4" name="interest"><label for="int4" style="width:auto;">Comics</label><br>

      <label for="genre">Preferred Genre</label>
      <select id="genre" name="genre">
        <option value="fiction">Fiction</option>
        <option value="nonfiction">Non-Fiction</option>
        <option value="scifi">Sci-Fi</option>
      </select><br>

      <label for="hcolor">Highlight Color</label>
      <input type="color" id="hcolor" name="hcolor" value="#ffd447"><br>

      <label for="permonth">Books per Month</label>
      <input type="range" id="permonth" name="permonth" min="1" max="10" value="4"><br>

      <label for="age">Age</label>
      <input type="number" id="age" name="age" min="10" max="100" value="21"><br>

      <label for="joindate">Membership Start</label>
      <input type="date" id="joindate" name="joindate"><br><br>

      <button type="submit">Register Membership</button>
    </form>
  </fieldset>
</div>

<div class="media">
  <h3>Orientation Media</h3>
  <audio src="../../../media/welcome_audio.mp3" controls></audio>
  <br><br>
  <video src="../../../media/orientation_video.mp4" controls width="260" height="160"></video>
</div>

<footer>
  <h2>HAPPY READING</h2>
</footer>

</body>
</html>
```

```
!DOCTYPE html>
html lang="en">
head>
  meta charset="UTF-8">
  meta name="viewport" content="width=device-width, initial-scale=1.0">
  title>Bookify Library Portal</title>
  style>
    html { scroll-behavior: smooth; }
    body { font-family: 'Segoe UI', 'Tahome, Verdana, sans-serif; margin: 0; background:#f4f6f9; color:#1f2937; }
    header { background:#123c6e; color:#fff; padding:24px 16px; text-align:center; }
    header h1 { margin:0; letter-spacing:2px; }
    header .tagline { margin:6px 0 14px 0; font-size:15px; }
    header p.intro { max-width:820px; margin:0 auto; font-size:14px; line-height:1.6; text-align:left; }
    nav { background:#0d2c52; padding:10px 20px; }
    nav .navwrap { display:flex; justify-content:space-between; flex-wrap:wrap; max-width:900px; margin:0 auto; }
    nav ol, nav ul { margin:8px 0; padding-left:20px; }
    nav a { color:#ffd447; text-decoration:none; font-weight:600; }
    nav li { color:#e3e7eb; margin:4px 0; }
    article { max-width:900px; margin:20px auto; background:#fff; padding:20px; border-radius:8px; box-shadow:0 1px 4px rgba(0,0,0,.1); }
    article .artwrap { display:flex; align-items:flex-start; gap:18px; }
    article img { border-radius:8px; }
    section { max-width:900px; margin:20px auto; }
    table { width:100%; border-collapse:collapse; background:#fff; box-shadow:0 1px 4px rgba(0,0,0,.1); }
    thead { background:#123c6e; color:#fff; }
    th, td { padding:10px 12px; border:1px solid #dde3ea; text-align:left; font-size:14px; }
    .formbox { max-width:900px; margin:24px auto; background:#fff; padding:20px; border-radius:8px; box-shadow:0 1px 4px rgba(0,0,0,.1); }
    fieldset { border:1px solid #cbd9e1; border-radius:6px; padding:16px; }
    legend { font-weight:700; color:#123c6e; padding:0 8px; }
    .formbox label { display:inline-block; width:170px; font-size:14px; margin:8px 0; }
    .formbox input, .formbox select { margin:6px 0; }
    .media { max-width:900px; margin:24px auto; background:#fff; padding:20px; border-radius:8px; box-shadow:0 1px 4px rgba(0,0,0,.1); text-align:
    footer { background:#0d2c52; color:#fff; text-align:center; padding:20px; margin-top:30px; }
/style>
/head>
body>

header>
<h1>BOOKIFY</h1>
<p class="tagline">Digital Membership H<sup>2</sup>O of Knowledge <sub>(Est. Reading Cloud)</sub></p>
<p class="intro">
  Bookify is the community library management portal for the Riverside Public Library Network,
  launched to help members browse the catalogue, track borrowed titles, and manage their
  membership entirely online. The system currently serves over 3,000 registered readers across
  four branch locations and supports both physical and e-book lending. This page demonstrates the
  core HTML5 building blocks used to construct the member-facing dashboard.
</p>
/header>

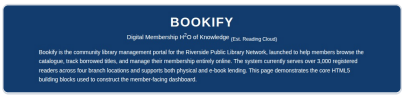
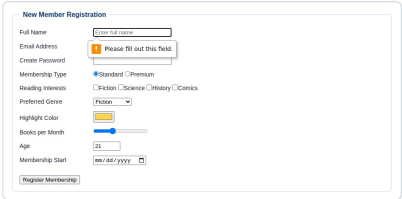
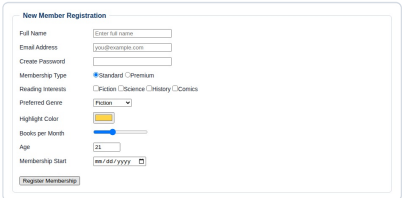
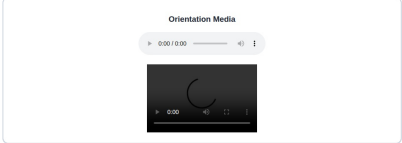
nav>
<div class="navwrap">
  <ol>
    <li>Dashboard</li>
    <li><a href="#catalogue">Catalogue Table</a></li>
    <li>My Loans</li>
    <li>Support</li>
  </ol>
  <ul>
    <li><a href="https://www.example-library.org/">Branch Locator</a></li>
    <li><a href="#registerForm">Join Now</a></li>
    <li>Events</li>
    <li>Careers</li>
  </ul>
</div>
/nav>

article>
<div class="artwrap">
  
  <p>
    Riverside Public Library was founded in 1994 as a single-branch reading room and has since
    grown into a four-branch network offering study pods, maker-space workshops, and a digital
    lending catalogue of over 45,000 titles. Bookify was built to give members a single dashboard
    for renewals, holds, and fine payments, reducing front-desk wait times by roughly sixty percent
    since its rollout last year.
  </p>
</div>
/article>
```

HTML Component / Tag	Functional Description
<header>, <nav>, <article>, <section>, <footer>	Semantic containers that define the banner, navigation bar, editorial content block, grouped section, and closing region of the page.
<sup> / <sub>	Render the tagline's "H2O" formula-style superscript and a subscript annotation with correct baseline offsets.
/	Provide two independently styled navigation lists — numbered dashboard links and bulleted portal/external links — including one internal hash-bookmark anchor.

<table>, <thead>, <tbody>	Organise the book-catalogue dataset into a structured row/column grid with a distinct header row.
<fieldset>, <legend>	Group the registration form's controls under a labelled, visually bordered section.
<input required>, minlength, min, max>	Provide native client-side constraint validation before the form can be submitted.
<input type='color' / 'range' / 'number' / 'date'>	HTML5 interactive controls providing a native colour picker, slider, numeric stepper, and date picker respectively.
<audio> / <video> controls	Embed native multimedia player nodes offering play/pause, seek, and volume controls without external plugins.

The test suite below documents verified execution of `library.html` in a Chromium-based browser (rendered via a Playwright automation script), including one negative test case that exposes a missing-validation defect.

TC ID	Test Description	Input / Action	Expected Result	Screenshot	Status
TC-01	Header, Sup/Sub Rendering	Render header section in Chromium	Site title displays with "H2O" superscript and subscript tagline at correct baseline offsets		Pass
TC-02	Two-Tier Navigation	Render nav bar in Chromium	Ordered dashboard list and unordered portal list display side-by-side with working anchor link		Pass
TC-03	Logo & Article Layout	Inspect article container	Library logo renders beside the descriptive history paragraph in a flex layout		Pass
TC-04	Catalogue Table	Inspect table data alignment	Book ID, Title, and Status columns align cleanly under the styled header row		Pass
TC-05	Required Field Validation	Submit form with Full Name & Email left blank	Browser must block submission and display a native "Please fill out this field" prompt		Pass
TC-06	Reading-Interest Constraint (Negative)	Fill Name & Email, submit with zero checkboxes selected	Form should require at least one interest to be checked; however, submission succeeds with no interests selected because no required/custom constraint was attached to the checkbox group		Fail
TC-07	Media Playback Nodes	Inspect audio & video player rendering	Native HTML5 audio and video elements render with visible play, seek, and volume controls		Pass

BOOKIFY

Digital Membership H²O of Knowledge (Est. Reading Cloud)

Bookify is the community library management portal for the Riverside Public Library Network, launched to help members browse the catalogue, track borrowed titles, and manage their membership entirely online. The system currently serves over 3,000 registered readers across four branch locations and supports both physical and e-book lending. This page demonstrates the core HTML5 building blocks used to construct the member-facing dashboard.

1. Dashboard

2. Catalogue Table

3. My Loans

4. Support

• Branch Locator

• Join Now

• Events

• Careers

BLP

Riverside Public Library was founded in 1994 as a single-branch reading room and has since grown into a four-branch network offering study pods, maker-space workshops, and a digital lending catalogue of over 45,000 titles. Bookify was built to give members a single dashboard for renewals, holds, and fine payments, reducing front-desk wait times by roughly sixty percent since its rollout last year.

Book ID	Title	Status
BK-1042	The Silent Orchard	Available
BK-2087	Vectors of Tomorrow	On Loan
BK-3391	Algorithms in the Attic	Reserved

New Member Registration

Full Name

Enter full name

Email Address

you@example.com

Create Password

Membership Type

☒ Standard ☐ Premium

Reading Interests

☐ Fiction ☐ Science ☐ History ☐ Comics

Preferred Genre

Fiction

Highlight Color

Books per Month

Age

21

Membership Start

mm / dd / yyyy

Register Membership

Orientation Media

▶ 0:00 / 0:00

🔊 ⋮

▶ 0:00

🔊 📺 ⋮

HAPPY READING

Thus, a structured and semantically rich web page (Bookify Library Portal) utilising the fundamental building blocks of HTML and HTML5 — semantic containers, superscript/subscript text formatting, dual-format navigation lists, a tabular book catalogue, a fully validated HTML5 registration form, and native multimedia audio/video controls — was successfully designed, implemented, and verified through seven documented test cases, including one negative test case that identified a genuine missing-constraint defect in the checkbox group.

Submission Package Structure

Observation/ObTask1/

```
└─ Docs/
  └─ ObsWeek1.pdf
└─ Screenshots/
  └─ full_page.png
  └─ tc01_header.png
  └─ tc02_nav.png
  └─ tc03_article.png
  └─ tc04_table.png
  └─ tc05_required_validation.png
  └─ tc06_interests_novalidation.png
  └─ tc07_media.png
└─ html/
  └─ library.html
└─ media/
  └─ library_logo.png
  └─ welcome_audio.mp3
  └─ orientation_video.mp4
```